

Security Mechanisms

Secure Your Devices , your network and your data

“Don't let Personal Security To Chance”

Practical Activity

simple, hands-on cybersecurity basics you can actually *do* on a normal computer.

- **First: What beginners should protect**
- At a basic level, cybersecurity means protecting:
- Your **computer**
- Your **network / Wi-Fi**
- **Protecting Your Mobile Phone**
- Your **accounts**
- Your **data**

Tools Used

- **Windows cmd**
- **Wireshark Tool**
- **Windows Defender**
- **Firewall**

We'll start small and practical.

Activity #1

Check open ports on your computer (VERY basic)

Open ports can be entry points for attackers.

♦ On Windows

1. Press **Win + R**

2. Type:

`netstat -ano`

3. Press Enter

You'll see ports like 80, 443, etc.

- **LISTENING** = active service

- If you see many unknown ports → worth investigating later

👉 **Beginner tip: Don't close ports randomly yet. Just observe.**

Activity #2

Block an IP address (practical & useful)

Blocking IPs is one of the first real security actions.

- ◆ **Windows Firewall (easy way)**

1. Open **Windows Defender Firewall**
2. Click **Advanced settings**
3. Click **Inbound Rules**
4. Click **New Rule**
5. Choose **Custom**
6. Select **All programs**
7. Choose **Any protocol**

1. Under **Remote IP addresses**, select:

-  “These IP addresses”
- Add the IP you want to block (example: 192.168.1.100)

2. Choose **Block the connection**

3. Name it: Blocked Suspicious IP

 Done — IP is blocked.

Activity #3

Secure your Wi-Fi (important!)

Most attacks start from weak Wi-Fi.

◆ Do this now:

- Change router password
- Use **WPA2 or WPA3**
- Disable **WPS**
- Change default router login (admin/admin is dangerous)

Basic firewall concept (VERY simple)

Think of a firewall like a **security guard**:

- ✓ Allows trusted traffic
- ✗ Blocks suspicious traffic

Beginner rules:

- Keep firewall **ON** always
- Only allow apps you **recognize**
- Block anything you **don't understand yet**

Password security (no tools needed)

 BAD:

```
nginx
```

```
password123
```

```
admin123
```

 GOOD:

```
graphql
```

```
S@f3!Net#2026
```

Password security (no tools needed)

Beginner rule:

- Minimum 12 characters
- Mix of letters + numbers + symbols
- Never reuse passwords

 **Use a password manager later (Bitwarden is beginner-friendly).**

Spot phishing (very important skill)

Dangerous signs:

- “Your account will be locked!”
- Misspelled emails
- Strange links
- Attachments you didn’t expect

Safe move:

- Don’t click
- Verify sender
- Type website address manually

Antivirus basics

You don't need fancy tools.

✓ Windows Defender is enough for beginners

Just:

- Keep it updated
- Run full scan once a week

Very simple monitoring trick

Check who is connected to your network:

- Log into your router
- Look for **Connected Devices**
- If you see unknown devices → change Wi-Fi password

Assignment#1

Beginner practice challenge (try this)

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- ✓ check your IP Address and Identify the Class
- ✓ Asks Your Friend IP and send traffic using “ping” command
- ✓ Block one test IP, in “windows defender firewall” any unwanted/
Malicious user
- ✓ Change Wi-Fi password
- ✓ Run antivirus scan
- ✓ Check connected devices

If you do these 6 things → you already practiced cybersecurity